Sprint 1 Final Submission Document

1.Sprint Details

- **Sprint Number: 1**
- **Sprint Pod Name:** EduSphere – E-Learning Management System
- **Pod Members:**
  - Member 1 – Aniket Das
  - Member 2 – Anish Dey
  - Member 3 – Arya Mane
  - Member 4 – Aymaan Shabbir
  - Member 5 – Bidisha Talukdar
  - Member 6 – Disha Makal
- **Submission Date:** 18-06-2025

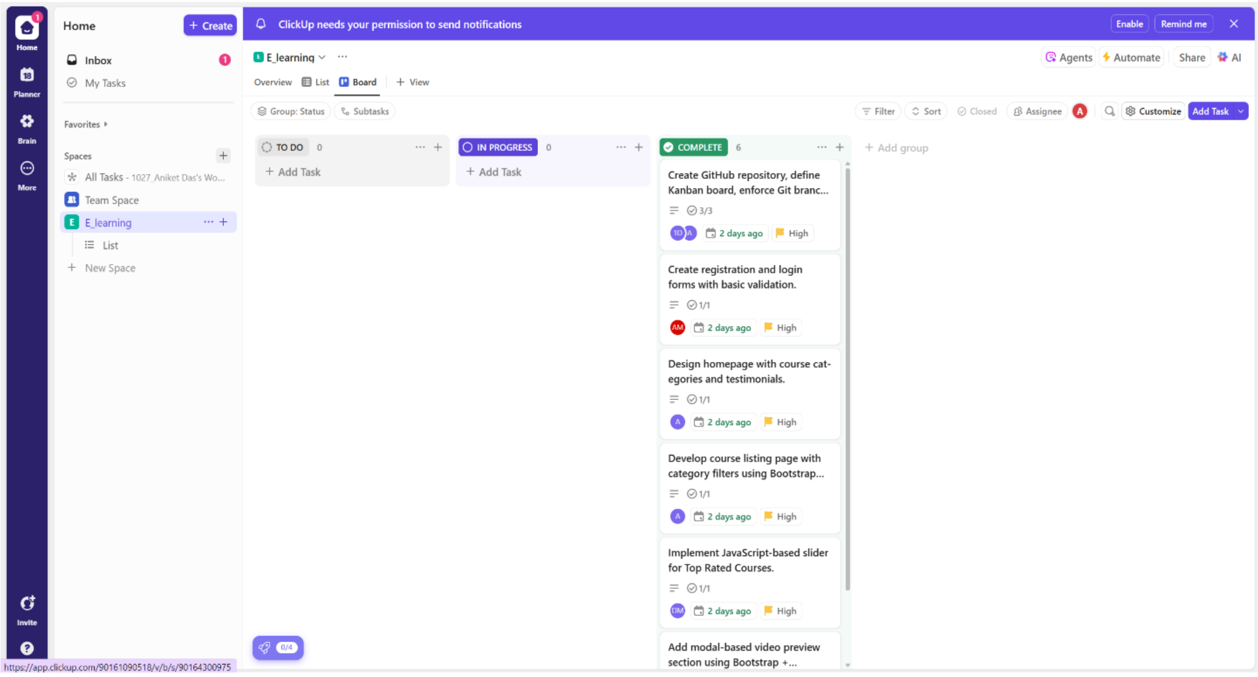
2. Sprint Goal

Set up a Git-based collaborative workflow and establish a visually appealing, responsive front-end foundation for EduSphere. This includes implementing core UI screens (homepage, login/registration, course listing), with interactivity and responsiveness using HTML5, CSS3, Bootstrap 5, and JavaScript.

3. User Stories Completion

User Story ID	Description	Assigned To	Status	Acceptance Criteria Met
US1	Create GitHub repository and enforce Git workflow	Member 1	Done	Yes
US2	Design homepage with course categories and testimonials	Member 2	Done	Yes
US3	Create registration and login forms with basic validation and roles	Member 3	Done	Yes
US4	Develop course listing page with category filters using Bootstrap grid	Member 4	Done	Yes
US5	Add modal-based video preview section using Bootstrap and JavaScript	Member 5	Done	Yes

US6	Implement JavaScript-based slider for “Top Rated Courses”	Member 6	Done	Yes



4. Code Submission

GitHub Repository Link: https://github.com/DISHA-6/E_Learning_System

Included Features:

- Homepage UI with Bootstrap layout
- Login and Registration pages with form validation
- Role-based navigation for Admin and User
- Course listing with category-based filtering
- Modal-based video previews
- JavaScript slider for top-rated courses

5. Sprint Review Document

What Was Planned:

In Sprint 1, we aimed to establish the **UI foundation** for our EduSphere E-Learning Management System. The primary goals included:

- Creating a centralized GitHub repository with a branching model
- Designing a responsive homepage with course categories and testimonials
- Developing login and registration pages with basic form validation
- Introducing role-based functionality for "Admin" and "User"
- Creating a course listing UI with filters using Bootstrap grid
- Adding interactivity like modal-based video previews and a slider for top-rated courses

What Was Delivered:

All the planned UI components and interactions were successfully developed and integrated. The completed deliverables include:

- **Responsive homepage** with:
  - Course categories section
  - Testimonials carousel
- **Login & Registration pages**:
  - With form validations (email format, strong password)
  - Role selection during registration (Admin/User)
- **Role-based UI rendering**:
  - Admin sees “Add Course” functionality
  - User sees enrolled course listings
- **Course Listing Page**:
  - Responsive grid using Bootstrap
  - Category-based filtering
- **Modal-based Video Preview**:
  - Each course card includes a preview button that opens a modal
- **Top Rated Courses Slider**:
  - JavaScript-powered slider displays featured courses

Key Limitation:

While Admins can access the "Add Course" interface, the new courses are not retained or visible to users after a page refresh. This is due to the lack of backend integration or persistent data storage.

We plan to resolve this in Sprint 2 using a local JSON file and REST API logic.

Challenges Faced:

1. **Role-Based Navigation Complexity**
   - Implementing conditional rendering purely in JavaScript required precise logic and DOM manipulation.
   - We had to simulate login flows for different roles manually due to lack of backend.
2. **Course Add Persistence**
   - We could not reflect Admin-added courses in the User view since there was no dynamic data connection.
   - The course data was static and hard-coded.
3. **Modal UI Interference**
   - When integrated with the Bootstrap layout, modal previews caused layout issues on smaller screens.
4. **Team Coordination**
   - Initial Git merge conflicts occurred due to inconsistent naming conventions and overlapping changes in shared files.

Key Learnings:

- We gained strong practical knowledge of responsive UI design using Bootstrap 5, including grids, modals, and carousels.
- Implemented JavaScript-based interactivity like sliders, modals, and form validations.
- Learned to simulate role-based functionality using only frontend logic.
- Understood the importance of planning data flow early, even in UI sprints.
- Strengthened our collaborative workflow using GitHub, pull requests, and a Kanban board.

6.Retrospective Notes

What Went Well:

- Clean UI design using Bootstrap
- Team collaborated well using Git and Kanban board
- Most features were tested and completed ahead of time

What Didn't Go Well:

- No dynamic storage of Admin-added courses
- Difficult to simulate data flow between roles without backend

Areas to Improve:

- Use a JSON file or temporary localStorage for testing flow
- Integrate mock API early in the process for dynamic data

Actionable Learnings for Next Sprint:

- Implement REST API and JSON integration for backend simulation
- Use modular TypeScript and component reusability
- Improve role-based routing with SPA techniques